

Exact Boundary Corrections for Thue–Morse Diffraction

Signed Stern polynomials, positive spectral extrapolation,
and sharp Sobolev limits

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Abstract

The finite Thue–Morse diffraction products approach a singular measure, but their individual Fourier coefficients do not merely have unspecified finite-size errors. We identify the entire error below the block-length cutoff with one signed Stern sequence. If $N = 2^m$, μ_m is the normalized squared modulus of the length- N Thue–Morse polynomial, and $\eta(k) = \hat{\mu}(k)$ is the limiting autocorrelation, then, exactly for $0 \leq k \leq N$,

$$\hat{\mu}_m(k) - \eta(k) = \frac{(-1)^m}{3N} B_k(-2), \quad \sum_{k \geq 1} B_k(-2) z^k = z \prod_{j \geq 0} (1 - z^{2^j})^2.$$

Consequently the positive probability density $\tilde{\mu}_m = (\mu_m + 2\mu_{m+1})/3$ reproduces every limiting Fourier coefficient through $|k| = N$ exactly. We obtain a distributional first correction, a convergent analytic finite-size expansion governed by two-colored binary partitions, alternating Poisson-kernel bounds, and sharp convergence rates in every admissible negative Sobolev space. The raw approximation has a critical logarithmic loss and a saturation rate N^{-1} ; the corrected approximation has the bandwidth-optimal rate $N^{\sigma-s}$, where $\sigma = \frac{1}{2} \log_2((1 + \sqrt{17})/4)$. An amplitude-parameter extension gives a sharp positivity threshold $a \geq -2/3$. All these statements are proved. Classical autocorrelation recurrences, Stern polynomials, and moment exponents are explicitly separated from the candidate-new finite-size synthesis. Accompanying code performs 19,037 exact rational checks.

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1 Scope, prior work, and the contribution

1.1 A particular frontier rather than another formula atlas

The supplied *Thue–Morse Atlas and Frontiers* [1] develops a large network of digital, algebraic, analytic, and Fabius–Rvachev identities. Its existing finite autocorrelation recurrences make the following question natural: *can the complete finite-block defect be isolated, rather than bounded, and then removed without losing positivity?* This article answers that question on the dyadic sequence of block lengths. The answer also identifies precisely the topology in which the isolated defect has a limit.

The object here is a measure on the frequency circle. It is not the compactly supported Rvachev density, a Fabius spline approximation, or the tail of a sinc product. The atlas’s cumulant corrections and geometric extrapolation for those objects therefore do not themselves give the results below. What is shared is a proof strategy: isolate a finite product, identify the defect algebraically, and only then estimate it in the appropriate function space.

The atlas source was inspected through the repository on 4 September 2026. The retrieved `.tex` blob has Git object identifier

7be75f3bc86f9c6712eec6d7550eaea53716434a.

This is a file-level provenance marker, not a claim that an older commit recorded inside that document is the current repository commit.

1.2 What is established background, and what is proposed as new

The Riesz product and the finite and infinite Fourier recurrences are established background; Baake and Grimm [2] explicitly give these recurrences and the singular-continuous limit. The polynomial family $B_k(t)$ is the Stern polynomial family introduced by Klavžar, Milutinović, and Petr [3]. Neither the family nor its specialization at $t = -2$ is claimed as a new sequence.

The eigenvalue $1 + \sqrt{17}$ and the associated correlation-dimension exponent are also not new. Zaks, Pikovsky, and Kurths [4] computed the correlation dimension; Mauduit, Montgomery, and Rivat [5] studied the even moments of Thue–Morse generating products. The explicit square-correlation sum used below also appears in the recent work of Coons, Mazáč, Pincus–Kazmar, and Stout [6]. We give short proofs to expose how these known energies control an additional finite-size object. General correlation background is developed in Baake and Coons [7].

The results presented as *candidate-new contributions* are the joint package of the following connections: the exact Stern boundary term for finite diffraction and forward correlations; its cancellation by a positive two-level product; the universal non-measure correction with sharp Sobolev convergence and saturation; the analytic defect factorization and its positive binary-partition coefficients; and the amplitude family’s sharp positivity threshold and corresponding rate law. Targeted searches and comparison with the supplied atlas did not locate this package or these exact formulations. This is a bounded literature assessment, not a certification of worldwide mathematical priority. The theorems themselves do not depend on a novelty claim.

Ingredient or conclusion	Status in this article
Riesz products and autocorrelation recurrences	Classical inputs, rederived to fix all normalizations.
Stern polynomials	Known family; used to identify the entire boundary defect.
Square-moment exponent	Known value; not advertised as a newly discovered dimension.
Positive Fourier-exact correction	Proved finite-size construction; main candidate-new bridge.
Sharp Sobolev error regimes	Proved application of exact defects and known-type energy transfer.
Analytic defect and partition coefficients	Proved factorization, positivity, and observable bounds.
Amplitude extension	Proved exactness, positivity threshold, and fixed-parameter rate theorem.
Lean status	Proof architecture only; no new Lean formalization is asserted.

2 Normalizations and the finite diffraction measures

2.1 The signed sequence and its two product decompositions

Write $s_2(n)$ for the number of ones in the binary expansion of n , and put

$$\varepsilon_n = (-1)^{s_2(n)}, \quad P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n, \quad N = 2^m. \quad (2.1)$$

The empty product has value one, so $P_0 = 1$. Binary concatenation gives

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n, \quad P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (2.2)$$

Indeed, choosing the term $-z^{2^j}$ in a subset of the factors records exactly the one-bits of the exponent. Both useful recursions follow:

$$P_{m+1}(z) = (1 - z)P_m(z^2) = (1 - z^N)P_m(z). \quad (2.3)$$

On the open disk $\mathbb{D}$, the infinite product

$$E(z) = \prod_{j=0}^{\infty} (1 - z^{2^j}) = \sum_{n=0}^{\infty} \varepsilon_n z^n \quad (2.4)$$

converges locally uniformly and has no zeros. For $|z| \leq r < 1$, $\sum_j |z|^{2^j} < \infty$, so the tail product converges to a nonzero value; no individual factor vanishes in $\mathbb{D}$. Coefficient stabilization identifies the Taylor coefficients with ε_n .

2.2 Fourier conventions

We use $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and

$$\widehat{u}(k) = \langle u, e^{-2\pi i k x} \rangle, \quad k \in \mathbb{Z}. \quad (2.5)$$

For a measure this means integration; for a periodic distribution it means its usual pairing with a smooth function. All the measures considered here are real and even, so their Fourier coefficients are real and even. Define

$$f_m(x) = \frac{1}{N} |P_m(e^{2\pi i x})|^2 = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)), \quad \mu_m = f_m(x) \, dx. \quad (2.6)$$

Orthogonality of the exponentials gives $\int f_m = 1$. Thus μ_m is a probability measure. Its Fourier coefficients are

$$r_m(k) := \hat{\mu}_m(k) = \begin{cases} \frac{1}{N} \sum_{j=0}^{N-k-1} \varepsilon_j \varepsilon_{j+k}, & 0 \leq k < N, \\ 0, & k \geq N, \end{cases} \quad r_m(-k) = r_m(k). \quad (2.7)$$

In particular, the coefficient at the endpoint $k = N$ is zero. Keeping this endpoint in later statements will be useful.

Lemma 2.1 (Finite Fourier recursion). *For $m \geq 0$ and $r \geq 0$,*

$$r_{m+1}(2r) = r_m(r), \quad r_{m+1}(2r+1) = -\frac{1}{2}(r_m(r) + r_m(r+1)). \quad (2.8)$$

The initial sequence is $r_0(k) = \mathbf{1}_{k=0}$.

Proof. Equation (2.3) gives $f_{m+1}(x) = (1 - \cos(2\pi x))f_m(2x)$. The Fourier frequencies of $f_m(2x)$ are even. Multiplication by $1 - (e^{2\pi i x} + e^{-2\pi i x})/2$ leaves even coefficients unchanged and creates each odd coefficient from its two neighbors. This is exactly (2.8). $\square$

Proposition 2.2 (The limiting measure). *The measures μ_m converge weakly to a unique probability measure μ . Writing $\eta(k) = \hat{\mu}(k)$ for $k \geq 0$, its coefficients satisfy*

$$\eta(0) = 1, \quad \eta(1) = -\frac{1}{3}, \quad (2.9)$$

$$\eta(2r) = \eta(r), \quad \eta(2r+1) = -\frac{1}{2}(\eta(r) + \eta(r+1)). \quad (2.10)$$

They are rational and obey $|\eta(k)| \leq 1/3$ for $k \geq 1$.

Proof. The recursion at $k = 1$ is $r_{m+1}(1) = -(1 + r_m(1))/2$, whence

$$r_m(1) = -\frac{1}{3} + \frac{1}{3} \left(-\frac{1}{2}\right)^m. \quad (2.11)$$

For every $k \geq 2$, the right sides in (2.8) involve smaller nonnegative indices. Induction on k therefore proves convergence of every fixed coefficient, with limits (2.9)–(2.10). Probabilities on the compact circle have weakly convergent subsequences. Every subsequential limit has these same Fourier coefficients. Trigonometric polynomials are uniformly dense in continuous functions, so those coefficients determine the limit uniquely and the entire sequence converges. Rationality and the bound follow by induction: for $k \geq 2$ each recursion either copies one previously bounded coefficient or takes minus the average of two such coefficients. $\square$

The measure μ is the classical singular-continuous Thue–Morse diffraction measure [2]. The results below require the elementary characterization just proved; singular continuity is only used when contrasting weak convergence with total-variation convergence.

3 The signed Stern sequence hidden in the boundary

Definition 3.1. Set $g(0) = 0$, and for $k \geq 1$ set

$$g(k) = \sum_{j=0}^{k-1} \varepsilon_j \varepsilon_{k-1-j}. \quad (3.1)$$

The shift by one makes the binary recursion particularly simple. Ordinary convolution of the series (2.4) gives

$$Q(z) := \sum_{k \geq 1} g(k) z^k = z E(z)^2. \quad (3.2)$$

Lemma 3.2 (Stern recursion and bounds). *The sequence in (3.1) satisfies*

$$g(1) = 1, \quad g(2r) = -2g(r), \quad g(2r+1) = g(r) + g(r+1). \quad (3.3)$$

The last recursion is an identity at $r = 0$ and a descending computation for $r \geq 1$. Moreover, for $k \geq 1$,

$$|g(k)| \leq k, \quad g(2^j) = (-2)^j, \quad \nu_2(g(k)) = \nu_2(k). \quad (3.4)$$

Here ν_2 of a nonzero negative integer means the valuation of its absolute value.

Proof. Let $c_n = \sum_{j=0}^n \varepsilon_j \varepsilon_{n-j}$, so $g(k) = c_{k-1}$. For an odd index $2r-1$, each term has one even and one odd subscript. Splitting by parity yields $c_{2r-1} = -2c_{r-1}$. For an even index $2r$, the even-even terms sum to c_r , and the odd-odd terms sum to c_{r-1} (with $c_{-1} = 0$). Thus $c_{2r} = c_r + c_{r-1}$, proving (3.3).

The bound follows because (3.1) is a sum of k signs. Iteration of the even recursion proves the power-of-two value. To prove the valuation statement, first show $g(k) \equiv k \pmod{2}$ by induction: even indices give an even value, while at an odd index the two smaller adjacent indices have opposite parity. Hence every odd-index value is odd. Writing $k = 2^j u$ with u odd now gives $g(k) = (-2)^j g(u)$ and the exact valuation. $\square$

The Stern polynomials [3] are defined by $B_0(t) = 0$, $B_1(t) = 1$, $B_{2r}(t) = tB_r(t)$, and $B_{2r+1}(t) = B_r(t) + B_{r+1}(t)$. Therefore

$$g(k) = B_k(-2). \quad (3.5)$$

A useful binary evaluation scheme keeps two adjacent values. For $v(k) = (g(k), g(k+1))^T$,

$$v(2k) = \begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix} v(k), \quad v(2k+1) = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} v(k), \quad v(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3.6)$$

Thus a single value can be computed using a number of small matrix updates linear in the bit length of k .

k	1	2	3	4	5	6	7	8
$g(k)$	1	-2	-1	4	-3	2	3	-8
$3\eta(k)$	-1	-1	1	-1	0	1	0	-1

k	9	10	11	12	13	14	15	16
$g(k)$	1	6	-1	-4	5	-6	-5	16
$3\eta(k)$	1/2	0	-1/2	1	-1/2	0	1/2	-1

4 An exact finite-block error, including the endpoint

Theorem 4.1 (Exact Stern boundary correction). *For $m \geq 0$, $N = 2^m$, and $0 \leq k \leq N$,*

$$r_m(k) = \eta(k) + \frac{(-1)^m}{3N} g(k). \quad (4.1)$$

In particular,

$$|r_m(k) - \eta(k)| \leq \frac{k}{3N}, \quad (4.2)$$

with equality whenever k is a power of two in this interval.

Proof. At $m = 0$, the only indices are 0 and 1. Their identities are respectively $1 = 1$ and $0 = -1/3 + 1/3$. Suppose the theorem holds at level m . For an even index $2r \leq 2N$, the difference at the next level equals the old difference at r . Since

$$\frac{(-1)^{m+1}}{6N} g(2r) = \frac{(-1)^m}{3N} g(r),$$

the claimed formula follows. For an odd index $2r + 1 \leq 2N$, both r and $r + 1$ lie in $[0, N]$. Subtracting (2.10) from (2.8) gives

$$-\frac{1}{2} \frac{(-1)^m}{3N} (g(r) + g(r + 1)) = \frac{(-1)^{m+1}}{6N} g(2r + 1).$$

This completes the induction, including the zero coefficient at the moving endpoint. Equation (3.4) proves the bound and its sharpness. $\square$

Corollary 4.2 (No nonzero stabilized harmonic is accidentally exact). *For every $k \geq 1$ and every m with $2^m \geq k$, $r_m(k) \neq \eta(k)$. For fixed k , the error changes sign with m and has exactly half the magnitude at the next level. Its rational two-adic valuation is $\nu_2(k) - m$.*

Proof. Use $g(k) \neq 0$ and its valuation in (4.1). The denominator factor 3 is odd. $\square$

The conclusion is about each fixed coefficient, not uniform convergence on the entire expanding band: at $k = N$, the error has magnitude $1/3$ for every m . The motion of the cutoff is exactly why function-space estimates later have a smoothness threshold.

4.1 Aperiodic versus forward correlations

The normalized coefficients r_m use a truncated sum, because both indices have to remain inside the polynomial block. The atlas also considers a forward sum whose second index may cross the right boundary. Define

$$C_m(k) = \sum_{j=0}^{N-k-1} \varepsilon_j \varepsilon_{j+k}, \quad A_m(k) = \sum_{j=0}^{N-1} \varepsilon_j \varepsilon_{j+k}, \quad 0 \leq k \leq N. \quad (4.3)$$

The sum defining $C_m(N)$ is empty.

Theorem 4.3 (Both finite-window conventions). *For $0 \leq k \leq N$,*

$$C_m(k) = N\eta(k) + \frac{(-1)^m}{3} g(k), \quad (4.4)$$

$$A_m(k) = N\eta(k) - \frac{2(-1)^m}{3} g(k). \quad (4.5)$$

Proof. The first formula is Theorem 4.1 multiplied by N . For the k terms crossing the right boundary, write $j = N - k + r$, $0 \leq r < k$. Binary complement inside an m -bit block gives $\varepsilon_{N-1-u} = (-1)^m \varepsilon_u$, and concatenation gives $\varepsilon_{N+r} = -\varepsilon_r$. Consequently

$$A_m(k) - C_m(k) = -(-1)^m \sum_{r=0}^{k-1} \varepsilon_{k-1-r} \varepsilon_r = -(-1)^m g(k).$$

Combining the two identities proves (4.5). $\square$

This theorem upgrades the finite autocorrelation recurrences to a closed finite-size expression valid simultaneously for every shift through the block length. In particular, the different constants $1/3$ and $-2/3$ are not conflicting normalizations: they correspond to distinct window conventions.

Example 4.4 (A length-four block). For $m = 2$, the block is $(1, -1, -1, 1)$ and

$$(r_2(0), \dots, r_2(4)) = (1, -1/4, -1/2, 1/4, 0).$$

Subtracting $(\eta(0), \dots, \eta(4))$ gives $(0, 1, -2, -1, 4)/12$, exactly the first five Stern values divided by $3N$.

5 Positive extrapolation with an exact Fourier band

Theorem 5.1 (The two-level positive correction). *For $m \geq 0$, define*

$$\tilde{\mu}_m = \frac{1}{3}\mu_m + \frac{2}{3}\mu_{m+1}. \quad (5.1)$$

It is a probability measure with nonnegative trigonometric-polynomial density

$$\tilde{f}_m(x) = f_m(x) \left(1 - \frac{2}{3} \cos(2\pi Nx)\right), \quad N = 2^m. \quad (5.2)$$

Its Fourier coefficients satisfy

$$\widehat{\tilde{\mu}_m}(k) = \eta(k) \quad (|k| \leq N). \quad (5.3)$$

Among affine combinations $w\mu_m + (1-w)\mu_{m+1}$, it is the unique one agreeing with the limit at $k = 1$. Its density has degree at most $2N - 1$.

Proof. The coefficient error at level $m + 1$ is $-1/2$ times the error at level m on the common band. The weights $1/3$ and $2/3$ cancel it exactly. The error at $k = 1$ is nonzero; hence any cancelling affine weight must solve $w - (1-w)/2 = 0$, which gives $w = 1/3$. The second factorization in (2.3) yields $f_{m+1} = f_m(1 - \cos(2\pi Nx))$, proving (5.2). Its new factor lies between $1/3$ and $5/3$. Positivity and total mass one also follow directly from the convex combination. The degree bound is the sum of $N - 1$ and N . $\square$

For every trigonometric polynomial ϕ of degree at most N , this gives an exact integration identity

$$\int_{\mathbb{T}} \phi \, d\mu = \int_0^1 \phi(x) \tilde{f}_m(x) \, dx. \quad (5.4)$$

Thus a singular measure has a positive, explicitly factored density representing its entire low-frequency moment vector. This is not an assertion that the two measures agree on arbitrary sets or arbitrary continuous functions.

5.1 Smooth and analytic observables

For $r \geq 0$, write

$$\mathcal{A}_r(\phi) = \sum_{k \in \mathbb{Z} \setminus \{0\}} |k|^r |\widehat{\phi}(k)|. \quad (5.5)$$

Every smooth periodic function has finite $\mathcal{A}_r$ for every fixed r : repeated integration by parts bounds its Fourier coefficients by any desired power.

Corollary 5.2 (Tail-only integration error). *For an absolutely convergent Fourier series,*

$$|\langle \tilde{\mu}_m - \mu, \phi \rangle| \leq 2 \sum_{|k| > N} |\widehat{\phi}(k)|. \quad (5.6)$$

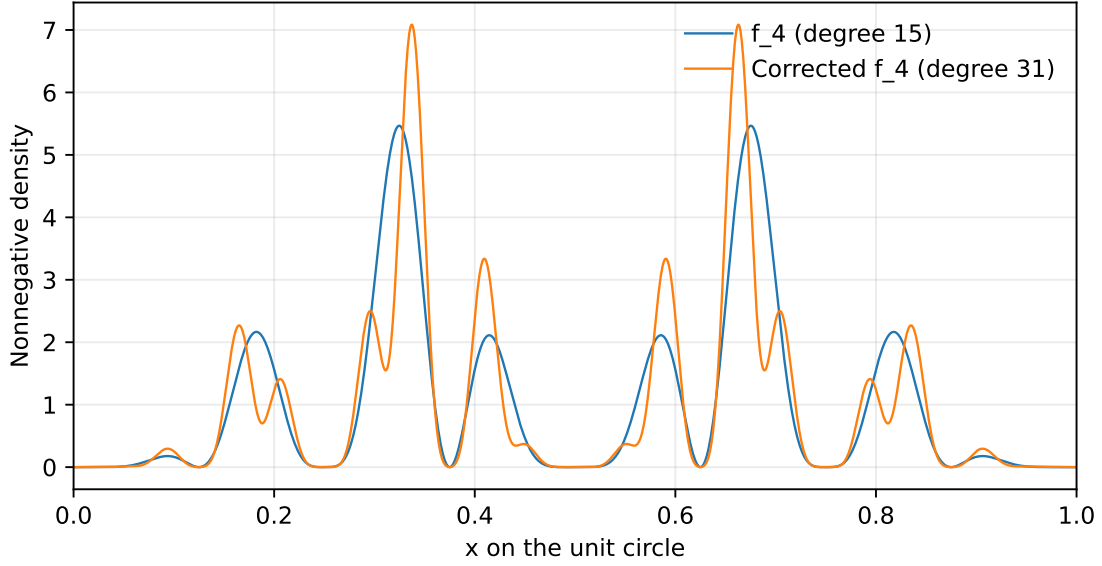


Figure 1: The raw and corrected densities at $m = 4$. Both are nonnegative. The corrected curve reproduces the limiting measure's Fourier coefficients through $|k| = 16$ exactly. The figure makes no assertion of pointwise convergence to a density of the limit.

Consequently, whenever $\mathcal{A}_r(\phi) < \infty$,

$$|\langle \tilde{\mu}_m - \mu, \phi \rangle| \leq 2N^{-r} \mathcal{A}_r(\phi). \quad (5.7)$$

If $|\hat{\phi}(k)| \leq M\rho^{|k|}$ for $0 < \rho < 1$, then

$$|\langle \tilde{\mu}_m - \mu, \phi \rangle| \leq \frac{4M\rho^{N+1}}{1 - \rho}. \quad (5.8)$$

Proof. Termwise pairing is justified by absolute convergence and finite total variation. Every coefficient on the low band vanishes by (5.3); every other coefficient of the difference has magnitude at most two because both measures are probabilities. Summing the Fourier tail gives (5.6); comparison with $|k|^r/N^r$ and a geometric series gives the other two statements. $\square$

For a fixed C^∞ observable, the corrected error decays faster than every power of N^{-1} . For a fixed analytic strip it decays exponentially in N . The constants depend on the observable or the strip, so these statements must not be applied uniformly to probes whose frequencies grow with N .

Remark 5.3 (Why positivity is a genuine constraint). Ordinary Richardson extrapolation often uses a negative weight. Here the raw error alternates in m , so the extrapolation weights are both positive. It is therefore exact on an expanding spectral band without sacrificing the measure interpretation. This does not imply convergence in total variation: since μ is singular and each $\tilde{\mu}_m$ is absolutely continuous, their total-variation norm distance is two, or one under the convention $\sup_A |\tilde{\mu}_m(A) - \mu(A)|$.

6 The universal first correction is a distribution, not a measure

Definition 6.1. Let ν be the real even periodic distribution specified by

$$\hat{\nu}(0) = 0, \quad \hat{\nu}(k) = \frac{1}{3}g(|k|) \quad (k \neq 0). \quad (6.1)$$

The bound $|g(k)| \leq k$ makes this definition legitimate: pairing its Fourier coefficients with those of a smooth function is absolutely convergent and continuous in an appropriate smooth seminorm.

Theorem 6.2 (One algebraic correction, with a rapidly small weak remainder). *There are distributions R_m such that*

$$\mu_m = \mu + \frac{(-1)^m}{N} \nu + R_m, \quad \widehat{R}_m(k) = 0 \quad (|k| \leq N). \quad (6.2)$$

For $r \geq 1$ and $\mathcal{A}_r(\phi) < \infty$,

$$|\langle R_m, \phi \rangle| \leq \frac{4}{3} N^{-r} \mathcal{A}_r(\phi). \quad (6.3)$$

In particular $(-1)^m N(\mu_m - \mu) \rightarrow \nu$ in the space of periodic distributions. The distribution ν is not a finite signed measure.

Proof. Define R_m by (6.2). The low coefficients vanish by Theorem 4.1. For $|k| > N$ the finite product has no coefficient, so

$$\widehat{R}_m(k) = -\eta(|k|) - \frac{(-1)^m}{3N} g(|k|).$$

Its magnitude is at most $1 + |k|/(3N) \leq 4|k|/(3N)$. Hence

$$|\langle R_m, \phi \rangle| \leq \frac{4}{3N} \sum_{|k| > N} |k| |\widehat{\phi}(k)| \leq \frac{4}{3} N^{-r} \mathcal{A}_r(\phi),$$

where $|k| \leq N^{1-r} |k|^r$ on this tail. Taking any fixed $r > 1$ proves the renormalized distributional convergence. Finally $|\widehat{\nu}(2^j)| = 2^j/3 \rightarrow \infty$. Fourier coefficients of a finite signed measure are bounded by its total variation, so ν cannot be one. $\square$

For each fixed smooth observable the expansion has just one nonzero algebraic finite-size term. All subsequent powers can be assigned zero coefficients, with a remainder smaller than every prescribed power. This is a weak statement, not a pointwise expansion of f_m . The negative Sobolev spaces in Section 10 quantify exactly how much smoothness is required to promote it to norm convergence.

7 An exact analytic factorization of all remaining modes

7.1 The affine Mahler equation and its homogeneous defect

Define the one-sided Fourier generating functions

$$H(z) = \sum_{k=0}^{\infty} \eta(k) z^k, \quad H_m(z) = \sum_{k=0}^{N-1} r_m(k) z^k, \quad |z| < 1. \quad (7.1)$$

These are not the same functions as E and P_m : E and P_m encode the sign sequence, while H and H_m encode autocorrelations.

Lemma 7.1 (Affine functional equations). *Writing $A(z) = 1 - (z + z^{-1})/2 = -(1 - z)^2/(2z)$, one has*

$$H(z) = A(z)H(z^2) + \frac{1}{2z}, \quad H_{m+1}(z) = A(z)H_m(z^2) + \frac{1}{2z}. \quad (7.2)$$

The singularities displayed at zero are removable.

Proof. The even part of either series is the previous series at z^2 . The odd part is

$$-\frac{z}{2} \left(H(z^2) + \frac{H(z^2) - 1}{z^2} \right)$$

for the limiting series, and the identical expression with H_m for a finite one. Adding the even part proves the equations. The constant coefficient is one on both sides, which also exhibits the cancellation at zero. $\square$

Theorem 7.2 (Exact analytic defect). *The function*

$$K(w) = \frac{3(1 - H(w))}{wE(w)^2}, \quad K(0) = 1, \quad (7.3)$$

is analytic on $\mathbb{D}$. For every $m \geq 0$ and $z \in \mathbb{D}$,

$$H_m(z) - H(z) = \frac{(-1)^m}{3N} zE(z)^2 K(z^N). \quad (7.4)$$

If $\tilde{H}_m = (H_m + 2H_{m+1})/3$, then

$$\tilde{H}_m(z) - H(z) = \frac{(-1)^m}{9N} zE(z)^2 (K(z^N) - K(z^{2N})). \quad (7.5)$$

Proof. Subtract the two equations in (7.2). Iterating the resulting homogeneous equation from $H_0 = 1$ gives

$$\begin{aligned} H_m(z) - H(z) &= \left(\prod_{j=0}^{m-1} A(z^{2^j}) \right) (1 - H(z^N)) \\ &= \frac{(-1)^m}{N} z^{1-N} P_m(z)^2 (1 - H(z^N)). \end{aligned}$$

The infinite product splits as $E(z) = P_m(z)E(z^N)$. Substitute $1 - H(w) = wE(w)^2 K(w)/3$ to obtain (7.4). The denominator in (7.3) has no zeros except its explicit first-order zero at zero, while $1 - H(w) = w/3 + O(w^2)$, so K is analytic and $K(0) = 1$. Combining two consecutive levels gives (7.5). $\square$

The proof also supplies a second derivation of the low-band theorem: in $Q(z)K(z^N)$, the nonconstant terms of K can contribute only from degree $N + 1$ onward. The exact boundary correction is precisely the constant term $K(0) = 1$.

7.2 Two-colored binary partitions and positive defect coefficients

Let $p_2(n)$ count partitions of n into powers of two, with each part available in two distinguishable colors. Equivalently,

$$\mathcal{P}(z) = \frac{1}{E(z)^2} = \prod_{j \geq 0} (1 - z^{2^j})^{-2} = \sum_{n \geq 0} p_2(n) z^n. \quad (7.6)$$

This product converges locally uniformly on the disk. The combinatorial interpretation follows by assigning independently a nonnegative multiplicity to each color of each power of two.

Theorem 7.3 (Finite coefficient formula and positivity). *The analytic function K satisfies*

$$2K(z) + K(z^2) = 3\mathcal{P}(z). \quad (7.7)$$

Writing $K(z) = \sum_{n \geq 0} \kappa_n z^n$, one has $\kappa_0 = 1$ and, for $n \geq 1$,

$$\kappa_n = \frac{3}{2} \sum_{j=0}^{\nu_2(n)} \left(-\frac{1}{2} \right)^j p_2(n/2^j). \quad (7.8)$$

All κ_n are positive, and

$$\frac{3}{4} p_2(n) \leq \kappa_n \leq \frac{3}{2} p_2(n) \quad (n \geq 1). \quad (7.9)$$

Proof. From (7.2),

$$1 - H(z) = A(z)(1 - H(z^2)) + \frac{z}{2}.$$

Using $E(z) = (1 - z)E(z^2)$ and the definition of K transforms this into $K(z) = 3/(2E(z)^2) - K(z^2)/2$, proving (7.7). Comparison of coefficients gives

$$2\kappa_n + \mathbf{1}_{2|n}\kappa_{n/2} = 3p_2(n) \quad (n \geq 1).$$

The descending recursion ends at an odd integer; expanding it proves the finite formula. Adding a part of size one of a fixed color is an injection from the partitions of n into those of $n + 1$, so $p_2(n)$ is nondecreasing. In the alternating finite sum the magnitude of each term is at most half that of the previous term. It lies between its first term minus its second and its first term. Those bounds, multiplied by $3/2$, yield (7.9); if the second term is absent the upper bound is attained. $\square$

For reference, the first coefficients are

$$\begin{aligned} (p_2(n))_{n=0}^8 &= (1, 2, 5, 8, 16, 24, 40, 56, 88), \\ (\kappa_n)_{n=0}^8 &= (1, 3, 6, 12, 21, 36, 54, 84, 243/2). \end{aligned}$$

The κ_n need not be integers. Their positivity is particularly useful: an alternating expression involving binary partitions becomes an analytic function with strictly positive Taylor coefficients.

Corollary 7.4 (A convergent finite-size expansion). *For $|z| < 1$,*

$$H_m(z) - H(z) = \frac{(-1)^m}{3N} Q(z) \sum_{j=0}^{\infty} \kappa_j z^{jN}. \quad (7.10)$$

For fixed $0 < r < 1$ and $J \geq 0$, truncation after $j = J$ has error $O_{J,r}(N^{-1}r^{(J+1)N})$, uniformly for $|z| \leq r$. The first mode missed by the positive correction is exactly

$$\widehat{\mu}_m(N+1) - \eta(N+1) = \frac{(-1)^m}{3N}. \quad (7.11)$$

Proof. Expand the analytic function K in (7.4). On $|w| \leq r$, its Taylor remainder after J is bounded by $C_{J,r}|w|^{J+1}$; for example, factor out that power and use continuity of the resulting analytic remainder on the compact disk. Put $w = z^N$ and bound Q on $|z| \leq r$. Finally $Q(z) = z + O(z^2)$, and $K(z^N) - K(z^{2N}) = 3z^N + O(z^{2N})$. Extracting the coefficient of z^{N+1} in (7.5) proves (7.11). $\square$

This is a convergent analytic expansion in the exponentially small variable z^N , multiplied by the alternating algebraic factor N^{-1} . It may be viewed as a finite-size transseries on compact subsets of the disk. No assertion of convergence on $|z| = 1$, nor an interchange with a boundary limit, is included. The distinction is essential because the circle supports the singular limiting measure and the non-measure correction.

8 Alternating bounds for Poisson observables

For $0 < \rho < 1$, consider the positive Poisson kernel centered at zero,

$$\Pi_\rho(x) = \frac{1 - \rho^2}{1 - 2\rho \cos(2\pi x) + \rho^2} = 1 + 2 \sum_{k \geq 1} \rho^k \cos(2\pi kx). \quad (8.1)$$

Its expectation under the limiting measure is $2H(\rho) - 1$. The positivity of K turns the analytic factorization into a signed, not merely absolute, error certificate.

m	N	Raw absolute error	Corrected absolute error
0	1	7.849×10^{-1}	2.5157×10^{-1}
1	2	1.5101×10^{-2}	4.4344×10^{-3}
2	4	8.9896×10^{-4}	2.0776×10^{-4}
3	8	1.3784×10^{-4}	1.7373×10^{-5}
4	16	4.286×10^{-5}	1.1421×10^{-6}
5	32	1.9717×10^{-5}	1.5597×10^{-8}
6	64	9.835×10^{-6}	6.1735×10^{-12}
7	128	4.9175×10^{-6}	1.9376×10^{-18}
8	256	2.4587×10^{-6}	3.8172×10^{-31}
9	512	1.2294×10^{-6}	2.9632×10^{-56}

Table 1: Absolute errors for $\Pi_{4/5}$. Values were evaluated with 160-digit arithmetic from the exact factorizations and independently checked against finite Laurent coefficients. Rounded decimal values are illustrations, not interval-arithmetic certificates. The signs are rigorously $(-1)^m$.

Theorem 8.1 (Poisson bracketing and strict improvement). *For every $m \geq 0$, the two errors $\langle \mu_m - \mu, \Pi_\rho \rangle$ and $\langle \tilde{\mu}_m - \mu, \Pi_\rho \rangle$ have sign $(-1)^m$ and are nonzero. Writing $t = \rho^N$, their ratio is*

$$\frac{\langle \tilde{\mu}_m - \mu, \Pi_\rho \rangle}{\langle \mu_m - \mu, \Pi_\rho \rangle} = \frac{K(t) - K(t^2)}{3K(t)} \in (0, 1/3). \quad (8.2)$$

As $m \rightarrow \infty$ this ratio is asymptotic to ρ^N . The even raw approximations decrease to the limit and the odd raw approximations increase to it. Both raw and corrected approximations give upper bounds at even levels and lower bounds at odd levels.

Proof. For real $\rho \in (0, 1)$, $Q(\rho) = \rho E(\rho)^2 > 0$. Theorems 7.2 and 7.3 give the exact errors

$$\langle \mu_m - \mu, \Pi_\rho \rangle = \frac{2(-1)^m}{3N} Q(\rho) K(t), \quad (8.3)$$

$$\langle \tilde{\mu}_m - \mu, \Pi_\rho \rangle = \frac{2(-1)^m}{9N} Q(\rho) (K(t) - K(t^2)). \quad (8.4)$$

Strict positivity of the coefficients implies $K(t) > K(t^2) > 0$. This proves all sign and ratio statements. Since $K(t) = 1 + 3t + O(t^2)$, the ratio equals $t + O(t^2)$. Finally the magnitude of the raw error at level $m + 2$, divided by that at level m , is $K(t^4)/(4K(t)) < 1/4$. Its sign is unchanged. This proves the nesting of each raw parity subsequence. $\square$

The theorem concerns Poisson kernels centered at zero. A spatially shifted kernel introduces oscillatory phases in its Fourier coefficients, so the same positivity argument does not automatically provide alternating upper and lower bounds. Likewise, nesting of the *corrected* parity subsequences is not needed and is not asserted here.

9 Exact dyadic energies and the classical moment exponent

9.1 A common two-dimensional transfer matrix

The growth of $g(k)$ along powers of two alone does not determine its Sobolev regularity. We need the total squared energy in a dyadic shell. Endpoint-weighted sums eliminate otherwise distracting boundary terms. Define

$$U_m = \sum_{k=1}^{N-1} g(k)^2 + \frac{1}{2} g(N)^2, \quad V_m = \sum_{k=0}^{N-1} g(k) g(k+1), \quad (9.1)$$

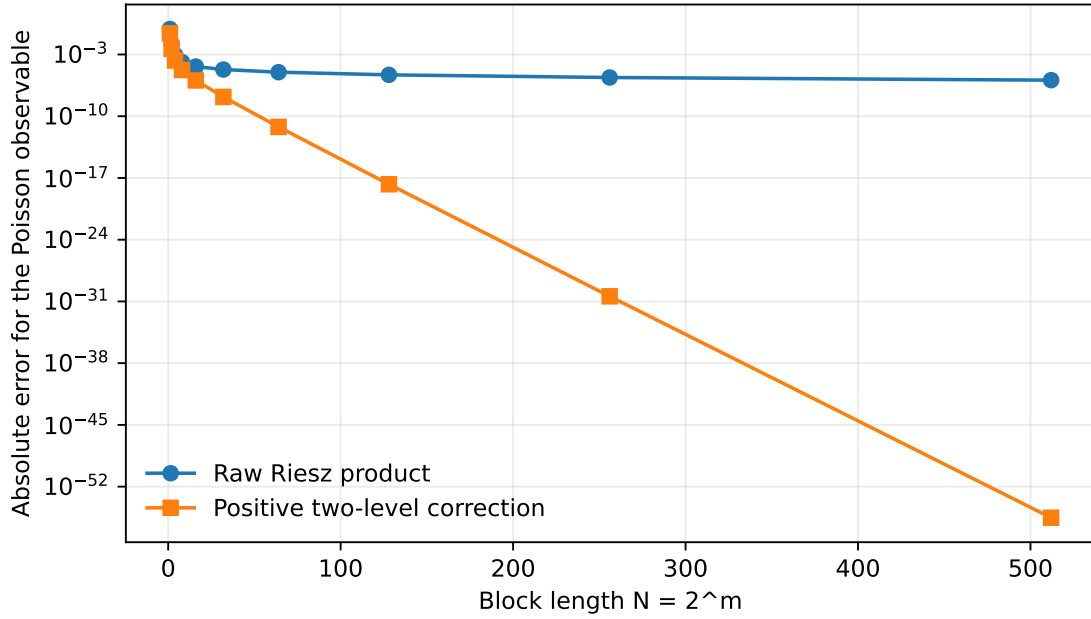


Figure 2: The same Poisson errors. The raw error eventually follows N^{-1} ; the corrected error has the additional factor $(4/5)^N$. This dramatic gain is for a fixed analytic observable, not a claim of total-variation convergence.

$$X_m = \sum_{k=1}^{N-1} \eta(k)^2 + \frac{5}{9}, \quad Y_m = \sum_{k=0}^{N-1} \eta(k)\eta(k+1). \quad (9.2)$$

The constant $5/9$ is the trapezoidal endpoint contribution $(\eta(0)^2 + \eta(N)^2)/2 = (1 + 1/9)/2$.

Proposition 9.1 (Exact energy recurrences). *Let*

$$M = \begin{pmatrix} 6 & 2 \\ -4 & -4 \end{pmatrix}. \quad (9.3)$$

Then

$$\begin{pmatrix} U_{m+1} \\ V_{m+1} \end{pmatrix} = M \begin{pmatrix} U_m \\ V_m \end{pmatrix}, \quad \begin{pmatrix} X_{m+1} \\ Y_{m+1} \end{pmatrix} = \frac{1}{4} M \begin{pmatrix} X_m \\ Y_m \end{pmatrix}, \quad (9.4)$$

with $(U_0, V_0) = (1/2, 0)$ *and* $(X_0, Y_0) = (5/9, -1/3)$.

Proof. For g , the even-index contribution to U_{m+1} is $4U_m$, including the half-weighted endpoint. The odd-index contribution is

$$\sum_{r=0}^{N-1} (g(r) + g(r+1))^2 = 2U_m + 2V_m,$$

because $g(0) = 0$. Thus $U_{m+1} = 6U_m + 2V_m$. Adjacent pairs at the two parities contribute

$$\sum_{r=0}^{N-1} [-2g(r)(g(r) + g(r+1)) - 2(g(r) + g(r+1))g(r+1)] = -4U_m - 4V_m.$$

This proves the first matrix equation.

For η , the even contribution to the trapezoidal square sum is X_m . The odd contribution is $(2X_m + 2Y_m)/4$, giving $X_{m+1} = (3X_m + Y_m)/2$. The adjacent-pair sum is $-(2X_m + 2Y_m)/2 = -X_m - Y_m$. This is precisely $M/4$. The initial values follow from $N = 1$, $g(1) = 1$, and $\eta(1) = -1/3$. $\square$

Corollary 9.2 (Closed forms). *Put $d = \sqrt{17}$, $\Lambda = 1 + d$, and $q = \Lambda/4$. Then*

$$U_m = \frac{5+d}{4d}\Lambda^m + \frac{d-5}{4d}(1-d)^m, \quad (9.5)$$

$$X_m = \frac{19+5d}{18d}q^m + \frac{5d-19}{18d}\left(\frac{1-d}{4}\right)^m. \quad (9.6)$$

In particular, $U_m \asymp \Lambda^m$ and $X_m \asymp q^m$.

Proof. The characteristic polynomial of M is $t^2 - 2t - 16$. Hence $U_{m+2} = 2U_{m+1} + 16U_m$ with $U_0 = 1/2, U_1 = 3$. For $M/4$ it is $t^2 - t/2 - 1$, and $X_0 = 5/9, X_1 = 2/3$. Solving these second-order recurrences gives the displayed formulas. The coefficients of the dominant positive roots are positive and the other roots have smaller modulus, proving the comparisons. $\square$

For clarity about prior work, $\sum_{k=0}^{N-1} \eta(k)^2 = X_m + 4/9$ is the square-correlation formula recorded in [6]. It is not a new exponent calculation. What matters here is that the boundary sequence has the *same* energy matrix, without the factor $1/4$.

9.2 The fourth moment as a Stern energy

Proposition 9.3. *For $N = 2^m$,*

$$\int_0^1 |P_m(e^{2\pi i x})|^4 dx = 2U_m, \quad \|f_m\|_{L^2}^2 = \frac{2U_m}{N^2}. \quad (9.7)$$

Proof. The coefficients of $P_m(z)^2$ through degree $N - 1$ equal $g(1), \dots, g(N)$, since truncating E at degree $N - 1$ cannot change these convolution coefficients. The reflection identity $z^{N-1}P_m(z^{-1}) = (-1)^m P_m(z)$ makes P_m^2 palindromic. Its coefficient squares therefore sum to $2 \sum_{k=1}^{N-1} g(k)^2 + g(N)^2 = 2U_m$. Parseval applied to P_m^2 proves the first equation, and division by N^2 proves the second. $\square$

This connects our transfer calculation to the classical even-moment setting of [5]. It also avoids numerical integration of highly oscillatory products.

9.3 Dyadic shell estimates

Define

$$\sigma = \frac{1}{2} \log_2 \frac{1 + \sqrt{17}}{4} = 0.178509318430288\dots, \quad s_* = 1 + \sigma = 1.178509318430288\dots \quad (9.8)$$

The familiar correlation dimension is $D_2 = 1 - 2\sigma = 3 - \log_2(1 + \sqrt{17})$ [4]. We will use only the energy estimates, not a new dimension theorem.

Lemma 9.4 (Two-sided shell energies). *With $N = 2^m$, constants independent of m give*

$$\sum_{N < k \leq 2N} g(k)^2 \asymp N^{2s_*}, \quad \sum_{N < k \leq 2N} \eta(k)^2 \asymp N^{2\sigma}. \quad (9.9)$$

Proof. Let $G_m = \sum_{k=1}^N g(k)^2 = U_m + 4^m/2$. Then the first shell sum is $G_{m+1} - G_m = U_{m+1} - U_m + 3 \cdot 4^m/2$. Since $\Lambda > 4$ and the dominant coefficient in (9.5) is positive, this is a positive constant times Λ^m asymptotically. The remaining finitely many shells have positive energy, so constants can be adjusted to cover them all. For η , the corresponding prefix sum through N is $X_m - 4/9$. Its shell difference is $X_{m+1} - X_m$, asymptotic to a positive constant times q^m , since $q > 1$. Each shell contains a power of two, where $\eta = -1/3$, so again every finite initial shell is nonzero. Finally $\Lambda^m = N^{2s_*}$ and $q^m = N^{2\sigma}$. $\square$

10 Sharp Sobolev regularity and finite-size convergence

10.1 The function spaces and a shell-summation lemma

For a periodic distribution u , use the norm

$$\|u\|_{H^{-s}}^2 = \sum_{k \in \mathbb{Z}} (1 + k^2)^{-s} |\widehat{u}(k)|^2. \quad (10.1)$$

The use of k rather than $2\pi k$ fixes constants but not thresholds. A pairing with $\phi \in H^s$ is bounded by $\|u\|_{H^{-s}} \|\phi\|_{H^s}$.

Lemma 10.1 (Weighted dyadic sums). *Suppose $a_k \geq 0$ and $\sum_{2^j < k \leq 2^{j+1}} a_k \asymp 2^{2tj}$, with a nonzero fixed low coefficient. For $N = 2^m$ and $s > t$,*

$$\sum_{k > N} (1 + k^2)^{-s} a_k \asymp N^{2(t-s)}. \quad (10.2)$$

The full weighted series converges if and only if $s > t$. Moreover, for positive s the prefix sum through N is, as $N \rightarrow \infty$,

$$\sum_{1 \leq k \leq N} (1 + k^2)^{-s} a_k \asymp \begin{cases} N^{2(t-s)}, & s < t, \\ \log N, & s = t, \\ 1, & s > t. \end{cases} \quad (10.3)$$

Proof. On the j th shell, the weight is comparable to 2^{-2sj} , with comparison constants depending only on s . The shell's weighted contribution is therefore comparable to $2^{2(t-s)j}$. Summing this geometric progression over $j \geq m$ gives (10.2); convergence fails at equality because each shell then contributes a positive amount. Summing over $j < m$ gives respectively a last-term-dominated progression, m comparable summands, or a bounded convergent series. The nonzero low coefficient gives the last case's positive lower bound. $\square$

Theorem 10.2 (Exact regularity thresholds). *The limiting measure and its boundary correction satisfy*

$$\mu \in H^{-s} \iff s > \sigma, \quad \nu \in H^{-s} \iff s > s_*. \quad (10.4)$$

Neither endpoint is included.

Proof. Apply Lemma 10.1 to the two shell estimates in Lemma 9.4. Evenness duplicates the positive frequencies, and the constant Fourier coefficient does not affect convergence. The factor $1/3$ in $\widehat{\nu}$ changes only the norm constant. $\square$

The difference of exactly one in the thresholds expresses a genuine derivative's worth of additional roughness in the universal correction. It is compatible with, but substantially more precise than, the pointwise bound $|g(k)| \leq k$.

10.2 Norm convergence of the renormalized error

Theorem 10.3 (Sharp topology for the first correction). *Set $Z_m = (-1)^m N(\mu_m - \mu)$. Then*

$$Z_m \longrightarrow \nu \text{ in } H^{-s} \iff s > s_*. \quad (10.5)$$

For $s > s_$,*

$$\|Z_m - \nu\|_{H^{-s}} \ll_s N^{s_* - s}. \quad (10.6)$$

Proof. On $|k| \leq N$ the difference vanishes. On the complement it equals $-(-1)^m N \eta(|k|) - g(|k|)/3$. The squared norm is bounded by a fixed constant times the sum of N^2 times the η tail energy and the g tail energy. By (10.2), both have size at most $N^{2(s_*-s)}$. This proves the norm bound and convergence. Conversely, a norm limit in H^{-s} must have every Fourier coefficient equal to the eventually exact coefficient $g(|k|)/3$, hence must be ν . That distribution belongs to H^{-s} only for $s > s_*$. For $s \leq \sigma$, already $\mu_m - \mu$ is not an element of H^{-s} ; the finite polynomial cannot remove the divergent tail. $\square$

10.3 The raw approximation: a phase transition and saturation

Theorem 10.4 (Sharp raw error rates). *For every fixed $s > \sigma$, as $N = 2^m \rightarrow \infty$,*

$$\|\mu_m - \mu\|_{H^{-s}} \asymp_s \begin{cases} N^{\sigma-s}, & \sigma < s < s_*, \\ N^{-1} \sqrt{\log N}, & s = s_*, \\ N^{-1}, & s > s_*. \end{cases} \quad (10.7)$$

For $s > s_$ there is also the exact leading norm limit*

$$N \|\mu_m - \mu\|_{H^{-s}} \longrightarrow \|\nu\|_{H^{-s}} > 0. \quad (10.8)$$

Proof. Because the finite coefficients vanish beyond their support, the squared norm has the exact decomposition

$$\|\mu_m - \mu\|_{H^{-s}}^2 = \frac{2}{9N^2} \sum_{k=1}^N (1+k^2)^{-s} g(k)^2 + 2 \sum_{k>N} (1+k^2)^{-s} \eta(k)^2. \quad (10.9)$$

The endpoint $k = N$ is included in the first sum by Theorem 4.1. Use (10.3) with $t = s_*$ for the first term and (10.2) with $t = \sigma$ for the second. If $s < s_*$, both are of order $N^{2(\sigma-s)}$. At $s = s_*$ the first is of order $N^{-2} \log N$, while the second is of order N^{-2} . Above s_* the first is of order N^{-2} and the second is smaller. Taking square roots proves the three regimes. Equation (10.8) follows from Theorem 10.3 and continuity of the norm; $\nu \neq 0$ because its first coefficient is $1/3$. $\square$

Thus additional smoothness ceases to improve the raw rate after s_* . The obstruction is not an estimate artifact: it is a specific nonzero distribution, with an exactly known Fourier series.

10.4 The corrected approximation removes saturation

Theorem 10.5 (Sharp corrected rate and bandwidth optimality). *For every fixed $s > \sigma$,*

$$\|\tilde{\mu}_m - \mu\|_{H^{-s}} \asymp_s N^{\sigma-s}. \quad (10.10)$$

More generally, fix $C > 0$. Every approximant v_N whose Fourier support is contained in $|k| \leq CN$ satisfies, for all sufficiently large dyadic N ,

$$\|v_N - \mu\|_{H^{-s}} \geq c_{s,C} N^{\sigma-s}. \quad (10.11)$$

No positivity assumption on v_N is required for this lower bound.

Proof. The corrected error vanishes through N . For the middle band $N < |k| < 2N$, use $(1+k^2)^{-s} \leq N^{-2s}$ and $|a-b|^2 \leq 2|a|^2 + 2|b|^2$. By (5.2) and Parseval,

$$\sum_k |\widehat{\tilde{\mu}_m}(k)|^2 = \|\tilde{f}_m\|_{L^2}^2 \leq \frac{25}{9} \|f_m\|_{L^2}^2 = \frac{50U_m}{9N^2} \ll N^{2\sigma}.$$

The η energy on the middle band has the same upper order by Lemma 9.4. Thus that band's weighted contribution is $O_s(N^{2(\sigma-s)})$. Beyond $2N$ the approximant has no coefficients, and Lemma 10.1 gives the same order for the tail. This proves the upper bound. The unchanged η tail beyond $2N$ alone proves the matching lower bound.

For the general assertion, choose a fixed integer j with $2^j > C$ and take the shell $2^j N < |k| \leq 2^{j+1} N$. All coefficients of v_N there vanish. Its weighted η energy is bounded below by a constant times $N^{2(\sigma-s)}$. Taking a square root proves (10.11). $\square$

For rough admissible tests, $\sigma < s < s_*$, the correction improves the constant and exactness band but cannot improve the order dictated by the unresolved spectral tail. At the threshold it removes $\sqrt{\log N}$, and above the threshold it removes the N^{-1} saturation completely. These are different regimes of a single exact identity, rather than competing convergence claims.

11 A parameter family and the sharp positivity threshold

11.1 Amplitude-deformed binary Riesz products

For $-1 \leq a \leq 1$, define

$$f_m^{(a)}(x) = \prod_{j=0}^{m-1} (1 - a \cos(2\pi 2^j x)), \quad \mu_m^{(a)} = f_m^{(a)}(x) dx. \quad (11.1)$$

These are amplitude deformations; no identification with a differently parameterized phase-deformation family is intended. Every factor is nonnegative. The constant coefficient remains one: in $f_{m+1}^{(a)} = (1 - a \cos(2\pi x))f_m^{(a)}(2x)$, the last factor has only even frequencies. Hence all $\mu_m^{(a)}$ are probabilities. Their Fourier recursion is

$$r_{m+1}^{(a)}(2r) = r_m^{(a)}(r), \quad r_{m+1}^{(a)}(2r+1) = -\frac{a}{2}(r_m^{(a)}(r) + r_m^{(a)}(r+1)). \quad (11.2)$$

The same fixed-index induction as in Proposition 2.2 proves a weak limit $\mu^{(a)}$, whose coefficients η_a obey this recursion with

$$\eta_a(0) = 1, \quad \eta_a(1) = -\frac{a}{2+a}. \quad (11.3)$$

The coefficient bound $|\eta_a(k)| \leq 1$ follows from the probability measure.

When $a = 0$, every measure is Lebesgue measure. When $a = 1$, we recover the Thue–Morse case. At $a = -1$,

$$f_m^{(-1)}(x) = \frac{1}{N} \left| \sum_{n=0}^{N-1} e^{2\pi i n x} \right|^2, \quad (11.4)$$

the dyadic Fejér kernel; its coefficients tend to one, so its limit is the point mass at zero. This endpoint is a useful exact check on the general formulas.

Theorem 11.1 (Parameterized Stern boundary). *Fix $a \in [-1, 1] \setminus \{0\}$, and write*

$$\alpha = -\frac{a}{2}, \quad c = \frac{a}{2+a}, \quad g_a(k) = B_k(-2/a). \quad (11.5)$$

Then, exactly for $0 \leq k \leq N = 2^m$,

$$r_m^{(a)}(k) - \eta_a(k) = c\alpha^m g_a(k). \quad (11.6)$$

Proof. The initial coefficient at $k = 1$ has error c , agreeing with $g_a(1) = 1$. The even recursion for g_a multiplies by $-2/a = \alpha^{-1}$, compensating for the extra α at the next level. Its odd recursion is the sum of two adjacent values, exactly matching the factor α in the Fourier recursion. The induction of Theorem 4.1, with these constants substituted, therefore proves the statement, including the endpoint. $\square$

Theorem 11.2 (Exactness and a sharp positivity interval). *For $a \in [-1, 1]$, put*

$$\tilde{\mu}_m^{(a)} = \frac{a\mu_m^{(a)} + 2\mu_{m+1}^{(a)}}{2+a}. \quad (11.7)$$

It has total mass one and agrees with $\mu^{(a)}$ at every Fourier mode $|k| \leq N$. Its density is

$$\tilde{f}_m^{(a)}(x) = f_m^{(a)}(x) \left(1 - \frac{2a}{2+a} \cos(2\pi Nx) \right). \quad (11.8)$$

For every m , this density is nonnegative on the whole circle if and only if

$$\boxed{-\frac{2}{3} \leq a \leq 1.} \quad (11.9)$$

For $a \neq 0$, the affine combination is uniquely determined by exactness at the first mode. At $a = 0$ all levels coincide and uniqueness is not asserted.

Proof. For $a \neq 0$, the common-band errors have ratio $\alpha = -a/2$. Their cancellation condition is $w + (1-w)\alpha = 0$, giving $w = a/(2+a)$. The density identity follows by factoring $f_{m+1}^{(a)} = f_m^{(a)}(1 - a \cos(2\pi Nx))$. The second factor is nonnegative exactly when $|2a/(2+a)| \leq 1$, which on $[-1, 1]$ is $a \geq -2/3$. This condition is sufficient since $f_m^{(a)} \geq 0$. If $a < -2/3$, the second factor is negative where $\cos(2\pi Nx) = -1$, for example at $x = 1/(2N)$: its value there is $(2+3a)/(2+a) < 0$. Every factor of $f_m^{(a)}$ at this point is strictly positive, since for $j < m$, $0 < 2\pi 2^j x \leq \pi/2$ and $1 - a \cos(2\pi 2^j x) \geq 1$. Thus the corrected density is strictly negative on a neighborhood of this point. This proves necessity, including the endpoint $a = -1$. The case $a = 0$ is immediate. $\square$

Remark 11.3 (Negative weights need not force a negative density). For $-2/3 \leq a < 0$, the coefficient of $\mu_m^{(a)}$ in the affine combination is negative, yet the resulting density is nonnegative. The factorization, rather than the signs of the mixing weights alone, decides positivity. Below $-2/3$ this exact two-level construction genuinely becomes signed. The theorem is a sharp threshold for this construction, not a no-go theorem for every possible positive approximation with a larger number of levels.

11.2 The full fixed-parameter Sobolev rate law

The same mechanism has a continuous family of exponents. The proof is included because the amplitude deformation separates two roles that coincide at $a = 1$: the boundary's geometric decay factor and the limiting measure's energy growth. Set

$$T_a = 1 - a + \frac{a^2}{2}, \quad \rho_a = \frac{T_a + \sqrt{T_a^2 + 4a}}{2}, \quad (11.10)$$

$$\beta_a = \log_2 \frac{2}{|a|}, \quad \sigma_a = \frac{1}{2} \log_2 \rho_a, \quad s_a = \sigma_a + \beta_a. \quad (11.11)$$

These quantities are defined for fixed $a \neq 0$ in $[-1, 1]$.

Theorem 11.4 (Parameterized regularity and sharp rates). *Let $\hat{\nu}_a(0) = 0$ and $\hat{\nu}_a(k) = c g_a(|k|)$ for $k \neq 0$, with c as in (11.5). Then*

$$\mu^{(a)} \in H^{-s} \iff s > \sigma_a, \quad \nu_a \in H^{-s} \iff s > s_a. \quad (11.12)$$

For every fixed $s > \sigma_a$,

$$\|\mu_m^{(a)} - \mu^{(a)}\|_{H^{-s}} \asymp_{a,s} \begin{cases} N^{\sigma_a - s}, & \sigma_a < s < s_a, \\ N^{-\beta_a} \sqrt{\log N}, & s = s_a, \\ N^{-\beta_a}, & s > s_a, \end{cases} \quad (11.13)$$

$$\|\tilde{\mu}_m^{(a)} - \mu^{(a)}\|_{H^{-s}} \asymp_{a,s} N^{\sigma_a - s}. \quad (11.14)$$

The corrected estimate remains valid below the positivity threshold, where the approximants are signed. For $s > s_a$, the renormalized errors converge to ν_a in H^{-s} after multiplication by α^{-m} , with error $O_{a,s}(N^{s_a - s})$.

Proof. Put $b = -2/a$ and define trapezoidal energies

$$\begin{aligned} U_m^{(a)} &= \sum_{k=1}^{N-1} g_a(k)^2 + \frac{1}{2}g_a(N)^2, & V_m^{(a)} &= \sum_{k=0}^{N-1} g_a(k)g_a(k+1), \\ X_m^{(a)} &= \sum_{k=1}^{N-1} \eta_a(k)^2 + \frac{1}{2}(1+c^2), & Y_m^{(a)} &= \sum_{k=0}^{N-1} \eta_a(k)\eta_a(k+1). \end{aligned}$$

The same even-odd splitting used in Proposition 9.1 gives

$$\begin{pmatrix} X_{m+1}^{(a)} \\ Y_{m+1}^{(a)} \end{pmatrix} = M_a \begin{pmatrix} X_m^{(a)} \\ Y_m^{(a)} \end{pmatrix}, \quad M_a = \begin{pmatrix} 1 + a^2/2 & a^2/2 \\ -a & -a \end{pmatrix}, \quad (11.15)$$

and

$$\begin{pmatrix} U_{m+1}^{(a)} \\ V_{m+1}^{(a)} \end{pmatrix} = \begin{pmatrix} b^2 + 2 & 2 \\ 2b & 2b \end{pmatrix} \begin{pmatrix} U_m^{(a)} \\ V_m^{(a)} \end{pmatrix} = \frac{4}{a^2} M_a \begin{pmatrix} U_m^{(a)} \\ V_m^{(a)} \end{pmatrix}. \quad (11.16)$$

For example the odd square contribution for g_a is $2U_m^{(a)} + 2V_m^{(a)}$; the adjacent-pair sum is $2b(U_m^{(a)} + V_m^{(a)})$. For η_a , the odd square contribution is $a^2(X_m^{(a)} + Y_m^{(a)})/2$, and the adjacent-pair sum is $-a(X_m^{(a)} + Y_m^{(a)})$. These establish both matrices without omitted boundary terms.

The characteristic polynomial of M_a is $x^2 - T_a x - a$. Its discriminant is positive. For $a \geq 0$ this is immediate. For $a = -u < 0$ it is $(1 - u)^2 + u^2 + u^3 + u^4/4 > 0$. Evaluating the polynomial at one gives $-a^2/2 < 0$, so the larger root ρ_a exceeds one. The other root has modulus $|a|/\rho_a < 1$. Thus each first-coordinate energy is a combination of two distinct real geometric sequences. The coefficient of ρ_a^m in $X_m^{(a)}$ is strictly positive: otherwise nonnegativity rules out a negative coefficient, while a zero coefficient would make the energy bounded, contrary to the m nonzero values $\eta_a(2^j) = -c$ in its prefix. For $U_m^{(a)}$, the larger eigenvalue is $(4/a^2)\rho_a > b^2$. The coefficient of that root is again positive: a zero dominant coefficient would contradict $U_m^{(a)} \geq b^{2m}/2$, because the smaller root has modulus strictly less than b^2 .

Consequently, taking differences of prefix energies as in Lemma 9.4,

$$\sum_{N < k \leq 2N} \eta_a(k)^2 \asymp_a N^{2\sigma_a}, \quad \sum_{N < k \leq 2N} g_a(k)^2 \asymp_a N^{2s_a}.$$

The endpoints $g_a(N)^2 = b^{2m}$ are lower order than the dominant energy; the endpoints $\eta_a(N)^2 = c^2$ are constant. All finite shells have positive energy because they contain powers of two. The sequence g_a also has polynomial growth: $|g_a(k)| \leq k^{\beta_a}$ follows by induction from $|b| = 2^{\beta_a}$ and $r^{\beta_a} + (r+1)^{\beta_a} \leq (2r+1)^{\beta_a}$, since $\beta_a \geq 1$. Thus ν_a is a distribution.

The shell-summation lemma now proves (11.12). On the low band the raw error is exactly $c\alpha^m g_a(k)$, with $|\alpha|^m = N^{-\beta_a}$; outside the support it is $-\eta_a(k)$. Applying the weighted prefix and tail estimates gives (11.13), including the logarithm at the critical exponent. The same tail argument proves the renormalized convergence and its rate.

For the corrected upper bound, its support is contained in $|k| < 2N$ and its error vanishes through N . The low-band identity also gives

$$\sum_k |r_m^{(a)}(k)|^2 \ll_a \sum_{k=0}^N \eta_a(k)^2 + |\alpha|^{2m} \sum_{k=0}^N g_a(k)^2 \ll_a N^{2\sigma_a}.$$

The multiplicative factor in (11.8) is bounded in absolute value by $1 + 2|a|/(2 + a)$. Hence its corrected L^2 norm has the same upper order, even when the factor changes sign. Bounding the middle band and the untouched tail exactly as in Theorem 10.5 proves the upper bound in (11.14). The tail beyond $2N$ proves the lower bound. $\square$

At $a = -1$, the formulas give $\rho_a = 2$, $\sigma_a = 1/2$, $\beta_a = 1$, and $s_a = 3/2$. Also $g_a(k) = B_k(2) = k$ and $c = -1$, so $r_m^{(-1)}(k) = 1 - k/N$ on $0 \leq k \leq N$, the familiar Fejér coefficients. At $a = 1$, every formula reduces to the Thue–Morse theorem. The limit $a \rightarrow 0$ is not uniform in the constants: the case $a = 0$ is exactly constant in m and is correctly excluded from the exponent parametrization.

12 Computation, verification, and proof engineering

12.1 Exact computation without constructing a large block

To compute a single stabilized finite Fourier coefficient, choose m with $2^m \geq k$, compute $\eta(k)$ from its two adjacent-value recursion, compute $g(k)$ using (3.6), and apply (4.1). The state for η has matrices

$$\begin{pmatrix} \eta(2k) \\ \eta(2k+1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1/2 & -1/2 \end{pmatrix} \begin{pmatrix} \eta(k) \\ \eta(k+1) \end{pmatrix}, \quad \begin{pmatrix} \eta(2k+1) \\ \eta(2k+2) \end{pmatrix} = \begin{pmatrix} -1/2 & -1/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta(k) \\ \eta(k+1) \end{pmatrix}. \quad (12.1)$$

Starting from $(\eta(0), \eta(1)) = (1, -1/3)$ and reading the bits from most to least significant takes $O(\log k)$ arithmetic operations. Leading zero bits leave this initial state unchanged. These are arithmetic-operation counts; bit complexity also accounts for integer sizes. The signed Stern values have $O(\log k)$ bits by $|g(k)| \leq k$, and the rational recursion for η grows its denominator by at most one factor of two per bit.

When all coefficients through N are required, the descending recurrences can be tabulated in $O(N)$ arithmetic operations and memory. Directly summing all correlations takes $O(N^2)$ sign multiplications; an FFT is an alternative for approximation, but the recurrence plus exact boundary term provides exact rational coefficients without rounding.

For the analytic correction, the finite formula (7.8) uses only $1 + \nu_2(n)$ binary partition values. A table of $p_2(0), \dots, p_2(L)$ can be computed by a coin-change dynamic program, using two colors of each size $1, 2, 4, \dots$, in $O(L \log L)$ integer additions. Formula (7.7) then computes all κ_n in $O(L)$ rational operations. Positivity prevents large cancellation when evaluating $K(t)$ for real $0 < t < 1$.

12.2 Avoiding loss of the corrected error

At large N , directly subtracting two nearly equal high-precision expectations can lose most significant digits. Equation (8.4) is a better evaluation formula. Even within that formula, write

$$K(t) - K(t^2) = \sum_{j \geq 1} \kappa_j t^j (1 - t^j), \quad 0 < t < 1, \quad (12.2)$$

instead of subtracting two numbers close to one. Every summand is positive. For very small t , one can factor out $3t$ before accumulating the remaining series. For complex z or shifted observables this particular positivity is lost, but the exact disk factorization still applies.

12.3 The supplied verification program

The companion `verify.py` uses only Python’s standard library. Identity checks use integers and `Fraction`; none uses floating-point equality. The recorded run performs 19,037 checks, including:

Test family	Range or independent representation
Aperiodic and forward corrections	Every $0 \leq k \leq 2^m$ for $0 \leq m \leq 10$.
Finite Fourier products	Independent Laurent-polynomial multiplication of all factors.
Positive correction and first missed mode	Exact low-band agreement and the coefficient at $N + 1$.
Signed Stern identity	Direct sign convolution through $k = 1024$, bounds, and valuations.
Energy transfer and fourth moments	Exact rational sums through level 12.
Analytic factorization	Every coefficient through degree 256 at levels 0 through 7.
Partition coefficient formula	Exact coefficients, finite sums, and positivity bounds through 256.
Amplitude family	Nine nonzero rational parameters, every low mode through level 7, and both energy matrices.

The parameters are $-1, -3/4, -2/3, -1/2, -1/7, 1/7, 1/4, 1/2, 1$. These tests are not replacements for proofs of infinite statements. They check independent finite representations and are particularly effective at catching endpoint, sign, normalization, and parameter mistakes.

`make_figures.py` separately uses `mpmath`, `numpy`, and `matplotlib`. The Poisson table uses 160 decimal digits and 2,201 terms for its independent limiting-series evaluation. The data and the exact test report are included as JSON. The verification program itself has no dependency on those optional plotting packages.

12.4 Formalization targets in the supplied repository

No Lean source is supplied or claimed checked here. The following dependency order isolates relatively small formal targets compatible with the atlas’s existing finite autocorrelation and polynomial-product development.

The first layer is entirely algebraic: define the convolution sequence g , prove its two parity recurrences, identify it with $B_k(-2)$ if a Stern-polynomial interface is desired, and prove the two exact boundary formulas. The latter need only natural-number induction, finite sums, binary reflection, and rational arithmetic. The central low-band cancellation is then a short consequence.

The second layer packages Fourier coefficients of finite trigonometric products. Positivity of the corrected density is a direct bound on cosine, not a limiting argument. Weak existence of μ can be taken from the existing development or proved from finite coefficient convergence and compactness.

The third layer separates analysis from combinatorics: prove the two energy matrix identities as finite-sum theorems, solve the scalar recurrences, derive shell estimates, and then apply an abstract weighted-shell lemma. The sharp Sobolev thresholds and all three error regimes become applications of that one lemma to an exact norm decomposition. The analytic factorization is a separate Mahler-equation branch; its only infinite-product requirements are local uniform convergence and nonvanishing in the open disk.

This architecture deliberately avoids equating “proved on paper” with “already formalized in the current repository.” It also keeps classical spectral-type results out of the dependencies of the new algebraic identities.

13 Further questions suggested by the exact mechanism

13.1 Non-dyadic windows and binary boundary states

For a length L that is not a power of two, the finite word decomposes into signed dyadic blocks. The single correction $g(k)$ should then be replaced by several cross-boundary states whose coefficients depend on the binary expansion of L . A concrete next problem is to derive a finite-dimensional exact

state system for these defects and determine which subsequences of L admit positive moment-exact combinations. The present theorems prove the one-state collapse on dyadic lengths; they do not assert such a collapse for all lengths.

13.2 Positive corrections for other substitution spectra

The general principle is an affine finite Fourier recursion with a contractive boundary eigenvalue. A negative eigenvalue allows cancellation by positive weights on two consecutive levels, as in the Thue–Morse case. With several boundary eigenvalues, the cancellation conditions become a finite moment problem for the level weights. Determining when these weights, or the resulting density factor, remain nonnegative is a concrete algebraic question. The interval $[-2/3, 1]$ in the amplitude family shows why positivity of the final factor can be a less restrictive condition than positivity of every weight.

13.3 Distribution functions and nonsmooth local probes

Theorems 10.4 and 10.5 describe an entire Sobolev scale, but they do not automatically settle optimal uniform error for cumulative distribution functions. Indicator functions are substantially less smooth than the observables for which the first correction gives a full weak expansion. A next step is to combine the exact band cancellation with sharp local mass bounds for the Thue–Morse measure. This requires additional local information, not another manipulation of the same fixed-mode identity.

13.4 More layers versus more exact modes

Two consecutive products uniquely cancel the one low-band boundary mode. Three or more levels can cancel selected later analytic sectors in (7.10), but the variables at consecutive levels are $z^N, z^{2N}, z^{4N}, \dots$, not ordinary powers of a common mesh parameter. A useful precise problem is to maximize the exact Fourier band subject to nonnegative level weights, or more generally a nonnegative factored density. The first missed coefficient (7.11) supplies the first nontrivial constraint. No general optimal multi-level scheme is claimed here.

14 Conclusion

The finite Thue–Morse diffraction defect is controlled by a simple object that is invisible in a statement of weak convergence alone: the signed Stern sequence $B_k(-2)$, equivalently the coefficients of $zE(z)^2$. On the full band $0 \leq k \leq 2^m$, it gives an exact, alternating boundary term. This one identity simultaneously explains the raw N^{-1} saturation for smooth tests, produces a positive Fourier-exact correction, and yields a universal rough distribution whose Sobolev threshold is determined by a classical moment eigenvalue.

The analytic factorization goes beyond the low band. It expresses every remaining finite-size sector through a single positive-coefficient function K , with explicit finite formulas in two-colored binary partition numbers. For Poisson observables this yields signed bounds and an exponential gain. The amplitude extension shows which parts of the mechanism persist away from Thue–Morse and precisely where positivity fails for the two-level construction. The resulting package is a self-contained extension of the supplied atlas, with proved formulas, sharp norm estimates, and reproducible exact checks rather than a list of numerical conjectures.

A Compact formula sheet

With $N = 2^m$, $g(k) = B_k(-2)$, and $Q(z) = zE(z)^2$, the central formulas are

$$\begin{aligned}
r_m(k) - \eta(k) &= \frac{(-1)^m}{3N} g(k), & 0 \leq k \leq N, \\
A_m(k) - N\eta(k) &= -\frac{2}{3} (-1)^m g(k), & 0 \leq k \leq N, \\
\tilde{\mu}_m &= (\mu_m + 2\mu_{m+1})/3, & \tilde{f}_m = f_m(1 - \frac{2}{3} \cos(2\pi Nx)), \\
\widehat{\mu}_m(k) &= \eta(k), & |k| \leq N, \\
\widehat{\nu}(k) &= g(|k|)/3, & k \neq 0, \\
H_m - H &= \frac{(-1)^m}{3N} Q(z)K(z^N), & |z| < 1, \\
2K(z) + K(z^2) &= 3E(z)^{-2}, & K(0) = 1, \\
\kappa_n &= \frac{3}{2} \sum_{j=0}^{\nu_2(n)} (-1/2)^j p_2(n/2^j), & n \geq 1.
\end{aligned}$$

The exponents are

$$\sigma = \frac{1}{2} \log_2 \frac{1 + \sqrt{17}}{4}, \quad s_* = 1 + \sigma.$$

The raw Sobolev rate is $N^{\sigma-s}$ below s_* , $N^{-1}\sqrt{\log N}$ at s_* , and N^{-1} above it. The corrected rate is $N^{\sigma-s}$ throughout $s > \sigma$. Every use of these rates fixes s before taking $N \rightarrow \infty$.

B Reproduction commands and file manifest

Run the exact checks independently of the document build:

```
python verify.py --output verification.json
```

The included report uses the default maximum direct-correlation level 10. A larger finite test can be requested with `-max-level 11`; direct correlation tests have quadratic cost in block length. Python 3.9 or later is sufficient for this standard-library program.

The figures and high-precision table are already included. To regenerate them, install the optional dependencies and run:

```
python -m pip install numpy matplotlib mpmath
python make_figures.py
```

The article uses a single main `.tex` file with an inline bibliography, a small generated table, and two figure PDFs. Build with a conventional TeX Live installation:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  Thue_Morse_Boundary_Corrections.tex
```

The preamble uses Libertinus when installed and falls back to Latin Modern. No font files are distributed. The ZIP includes the compiled article, its sources, the two reproducibility scripts, JSON data, rendered figure files, and a README.

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